

CTU-Chat: Supervisor-Routed Multi-Agent RAG with Heterogeneous Knowledge Allocation for University Counseling

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Abstract. University counseling spans relational curricula, cohort-specific fees, exemption policies, and narrative regulations, each requiring distinct evidence representations and computational treatments. Whether decomposing this heterogeneous task into multiple specialized agents improves end-to-end reliability over a single agent with equivalent capabilities remains underexplored. We present CTU-Chat, a supervisor-routed multi-agent system that assigns queries to four bounded specialists with scoped tool privileges and representation-specific evidence paths: Neo4j graph traversal for curricula, structured lookup with deterministic calculation for financial data, and hybrid document retrieval with neural reranking for regulations. We evaluate CTU-Chat through three experimental scenarios on a 100-query held-out benchmark: (1) an architecture-level comparison against a functionally matched single-agent baseline and a routed generic-worker variant under identical model, knowledge, and tool capabilities; (2) architectural ablations that individually remove specialist policies, tool isolation, heterogeneous evidence allocation, deterministic calculation, and route repair; and (3) a semantic-neighbor tool-ambiguity stress test across registries of 11, 31, and 51 tools. In architecture comparison, CTU-Chat obtains a 10.0 pp higher End-to-End Success point estimate (76.0% vs. 66.0%; paired 95% CI [+2.0, +19.0] pp; exact McNemar $p=0.0414$) and 41.4% fewer input tokens at the cost of higher latency ($\sim 2\times$). Under registry expansion to 51 tools, the scoped specialist configuration achieves 88.0% E2E execution (vs. 78.7% for Top-10 selection), without displaying statistically divergent degradation slopes. These findings suggest that domain-scoped specialization and heterogeneous evidence allocation can support institutional counseling reliability under the evaluated conditions, with measurable trade-offs in coordination latency and token cost.

Keywords: Multi-Agent Systems · Supervisor Routing · Architectural Ablation · Heterogeneous Knowledge Allocation · Retrieval-Augmented Generation · University Counseling

1 Introduction

Can Tho University (CTU) serves nearly 49,000 undergraduates in 2025–2026 [1]. Its scale and evolving regulations create advisory needs spanning heterogeneous knowledge forms: (1) **academic curricula** encoding hierarchies and prerequisites; (2) **tuition schedules** varying by cohort and field; (3) **exemption and scholarship policies** combining eligibility conditions with rates; and (4) **administrative regulations** comprising lengthy narrative documents. Prior systems address individual domains—CAAS for IT admission counseling [10] and REBot for regulation advising [9]—but a single representation or retrieval method may be unsuitable across all domains.

RAG and GraphRAG augment LLMs with documents and knowledge graphs [8,5,3], while agentic architectures support tool use and multi-agent coordination [20,18,16]. These approaches do not prescribe how heterogeneous institutional evidence should be represented, routed, and processed. Known limitations in language-model arithmetic [14] further motivate deterministic calculation for financial queries.

Central hypothesis. When a counseling system must handle diverse knowledge forms and semantically adjacent tools, routing queries to bounded specialists with representation-specific evidence paths may improve reliability relative to a single agent with equivalent capabilities, albeit with coordination latency. Whether this trade-off is favorable requires direct comparative evaluation.

These challenges motivate three gaps: (1) limited evidence comparing multi-agent and single-agent architectures under matched capabilities; (2) limited mechanistic understanding of which components drive observed differences; and (3) limited evidence on how bounded specialization maintains reliability as the tool registry grows. We formulate three research questions:

- **RQ1:** How does the proposed multi-agent architecture compare with a functionally matched single-agent in end-to-end reliability, evidence grounding, and cost?
- **RQ2:** What performance changes are observed when specialist policies, tool isolation, representation-specific evidence access, deterministic computation, and route repair are individually removed or replaced?
- **RQ3:** How does bounded specialist routing affect tool-selection reliability as semantically overlapping tools expand the registry from 11 to 51 tools?

This work provides three contributions: **C1**—a supervisor-routed bounded multi-agent architecture with four specialists, deterministic route repair, and partitioned tool privileges; **C2**—an architecture-level comparative evaluation against a functionally matched single-agent baseline and a routed generic-worker variant; and **C3**—architectural ablations of five components and a semantic-neighbor tool-registry stress test across 11–51 tools.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the CTU-Chat architecture. Section 4 describes the experimental setup. Section 5 reports results. Section 6 discusses findings and limitations. Section 7 concludes.

2 Related Work

2.1 Vietnamese NLP and Retrieval Foundations

Vietnamese is an isolating, tonal language where white spaces separate syllables rather than words [11]; 85% of word types contain at least two syllables. While domain-adapted bi-encoders capture semantic paraphrasing, Vietnamese higher-education decrees rely on rigid Sino-Vietnamese legal terminology and alphanumeric cohort codes, motivating coupling dense embeddings with exact lexical matching.

BM25 [15] provides reliable exact token matching for rigid identifiers but cannot capture paraphrases. Dense retrieval [7,8] captures paraphrastic intent but may struggle with rare alphanumeric tokens. Reciprocal Rank Fusion (RRF) [2] merges BM25 and dense candidates without score calibration, while neural cross-encoders [12,19] jointly encode query–passage pairs for reranking. Adaptive-RAG [6] demonstrates complexity-based retrieval selection. These approaches motivate dynamic retrieval but do not allocate institution-specific relational, tabular, and narrative knowledge to isolated specialist paths.

2.2 Knowledge Graphs and GraphRAG

Knowledge graphs represent domain knowledge as structured triples [13]. GraphRAG [3] leverages graph topology to uncover multi-hop associative paths that escape flat chunking, benefiting curriculum structures and prerequisite chains. However, universal graph extraction may introduce spurious relationships, motivating selective graph construction restricted to domains with well-defined relational schemas.

2.3 Agentic RAG and Multi-Agent Orchestration

ReAct [20] demonstrates interleaved reasoning and action, AutoGen [18] provides a multi-agent conversation framework, and Toolformer [16] studies tool invocation. For institutional counseling, these capabilities raise design risks without explicit constraints: broad tool exposure increases errors; peer-to-peer exchanges increase coordination cost; a shared index mixes evidence across domains. Importantly, controlled comparisons against functionally matched single-agent baselines remain limited, as most evaluations report absolute performance without isolating the contribution of multi-agent decomposition from underlying capabilities.

2.4 Educational Counseling Systems

Educational chatbots assist in advising and learning support [17]. CAAS supports admission counseling for IT programs [10], while REBot combines retrieval with category-guided GraphRAG for academic regulations [9]. CTU-CHAT departs from prior systems across three aspects: (1) *Task scope*: modeling 113 curricula as relational property graphs with deterministic arithmetic for tuition; (2) *Multi-agent routing*: a LangGraph supervisor routes to four bounded specialists with isolated evidence spaces; and (3) *Architectural evaluation*: direct comparison against functionally matched baselines with component-level ablations.

2.5 Research Gaps

Synthesizing the reviewed approaches identifies three gaps: (1) limited architecture-level evidence comparing multi-agent and single-agent systems under matched capabilities; (2) limited mechanistic understanding of which components drive observed differences; and (3) limited robustness evidence under tool ambiguity as semantic neighbors accumulate. These gaps motivate the evaluation design in Sections 4–5.

3 Proposed Model

This section presents CTU-CHAT’s architecture, algorithms, and evidence-acquisition mechanisms. The design follows three principles: *centralized agent coordination* (a contract-checked supervisor dispatches each query to one domain agent); *heterogeneous knowledge allocation* (each information structure maps to a corresponding evidence representation); and *representation-aware evidence routing* (graph-tool or intent-scoped retrieval path according to the assigned domain).

3.1 Overall Architecture

Figure 1 illustrates three coordinated tiers: (1) the Interaction and Centralized Supervisor Tier, (2) the Domain-Specialized Multi-Agent Core, and (3) the Heterogeneous Knowledge Storage and Tool Tier.

CTU-CHAT specializes LangGraph through typed state (`AgentState` with validated rewrite, routes, evidence, and response), an intent-owner contract that checks structured supervisor output before dispatch, conditional routing (academic queries go directly to the graph agent; others through retrieval), acyclic orchestration (specialists return to END without peer edges), and asymmetric tool registries (each specialist sees only its scoped tools, reducing per-query token cost and limiting the semantic-ambiguity surface as the tool registry scales).

3.2 Neo4j Knowledge Graph and Cypher Tools

The Neo4j graph is built from 114 Markdown source files covering 111 undergraduate curricula and fee schedules. The resulting graph stores approximately 7,000 nodes across 10 labels—curriculum entities (Program, Course, Block, PLO, PEO, SpecializationTrack, JobPosition) and tuition entities (TuitionFee, TuitionPolicy, ExemptionBasisRate)—connected by over 16,000 edges across 13 typed relationships encoding containment, prerequisites, program outcomes, tracks, occupations, tuition, and exemption bases.

The system does not ask the LLM to generate arbitrary Cypher. After the supervisor selects the academic agent, its ReAct policy chooses one of six typed tools and supplies arguments. Each tool normalizes identifiers and calls a fixed, parameterized Cypher query implementing three query classes: *attribute lookup/search* (program codes, names, comparisons), *structural traversal* (prerequisite chains up to ten hops, course intersection, reverse path search), and *structured financial lookup* (tuition, policies, exemption rates). This design binds agent actions to deterministic graph operations, reducing exposure to errors from unconstrained Cypher generation.

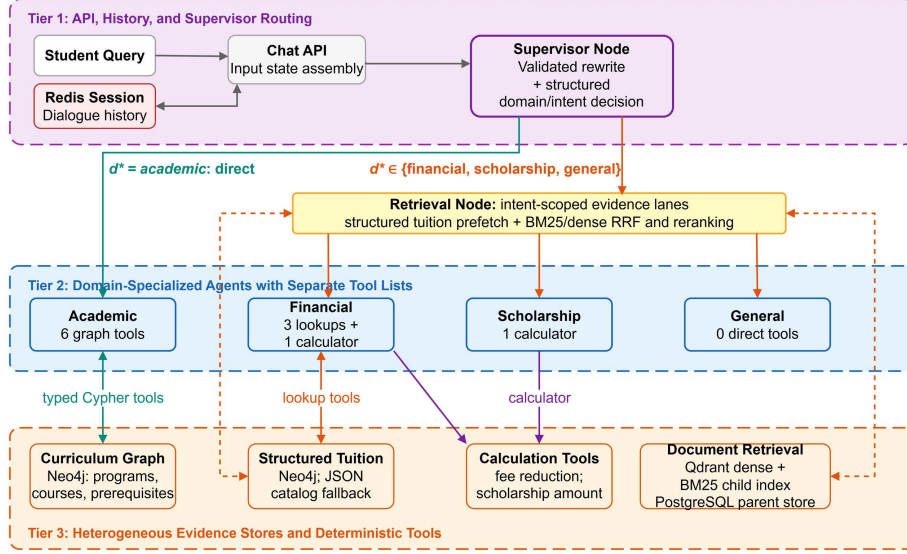


Fig. 1. Implemented CTU-CHAT architecture. Redis history enters the graph state; the supervisor’s structured decision is contract-checked before conditional dispatch. Academic queries enter the graph-enabled agent; other domains pass through intent-scoped retrieval.

3.3 Routing and Deterministic Arithmetic

For a query q and its dialogue predecessor, the supervisor first attempts a validated standalone rewrite, then predicts domain d_{raw} and intent t_{raw} under a structured schema, falling back to **general/other** on parsing failure. The ownership contract repairs rule conflicts or incompatible domain–intent pairs. Academic queries proceed directly to the graph agent; other domains build intent-scoped retrieval lanes, with financial queries first attempting Neo4j tuition lookup and then JSON fallback.

Deterministic Tuition Arithmetic. For fee-reduction requests, the tool `tin_h_toan_hoc_phi` receives three parameters—per-credit tuition T , per-credit exemption basis B , and percentage p —and applies:

$$D = B \frac{p}{100}, \quad P = \max(0, T - D). \quad (1)$$

The tool enforces $0 \leq p \leq 100$ and clamps P at zero. The contribution lies in composing validated rewriting, constrained routing, and deterministic arithmetic within the multi-agent workflow.

3.4 Heterogeneous Knowledge Allocation and Hybrid Retrieval

Rather than a uniform document index, CTU-CHAT preserves relational curricula in Neo4j, accesses cohort-dependent tuition through structured lookup, and retains narrative

regulations in document retrieval. Academic queries enter the graph agent directly; others enter retrieval. Financial retrieval first attempts Neo4j tuition lookup with JSON fallback; the repaired intent then selects metadata-filtered document lanes.

For narrative regulations and scholarship criteria, BM25 and Qdrant search 400-character child chunks; parent sections (2,800-char target, 100-char overlap) are loaded from PostgreSQL and fused via RRF: $s_{\text{RRF}}(d) = \sum_{s \in \{\text{bm25}, \text{dense}\}} (60 + r_s(d))^{-1}$. The full stack pools intent-governed and unfiltered candidates, deduplicates, and reranks with `bge-reranker-v2-m3`. Scores below 0.20 are removed unless fewer than two candidates remain; graph evidence is added under fixed quotas and up to seven passages are packed for generation.

3.5 Cross-Domain Query Handling

A bounded specialist architecture raises the question of how queries spanning multiple domains are handled. CTU-CHAT adopts a *single-owner with signal-based evidence prefetch* policy: the supervisor selects one primary owner based on dominant intent, and the retrieval layer expands evidence lanes when keyword signals indicate related domains—for example, adding academic-program or academic-regulation lanes when curriculum terms are detected, a tuition lane when fee-related terms appear, or a scholarship lane when scholarship keywords are present. Specialists do not invoke other specialists; if the query requires evidence outside the owner’s scope, the signal-detected lanes prefetch it through retrieval rather than cross-specialist delegation. This design trades full cross-domain tool composability for bounded execution with targeted evidence expansion. The 20 cross-domain held-out queries directly test this mechanism.

4 Experimental Setup

4.1 Dataset and Annotation

We evaluate CTU-CHAT on 200 administrative queries constructed through a document-grounded protocol over official CTU statutes. The 100-query development split supports index calibration and prompt engineering. The 100-query held-out set follows a predefined distribution: 40 direct, 20 within-domain multi-hop, 20 cross-domain, 10 entity-comparison, 5 temporal, and 5 adversarial queries (51 formal, 49 colloquial; 56 single-source, 44 multi-source). It contains 19 Academic, 28 Financial, 17 Scholarship, and 16 General single-domain queries; the remaining 20 combine evidence across domains. Candidate queries and ground-truth evidence were extracted from official statutes: one annotator established reference answers, required facts, and gold sources, while a second annotator independently verified all annotations; disagreements were adjudicated against official regulatory text. An audit found no cross-split duplicates (max 5-gram Jaccard = 0.1875). Each query is annotated along nine dimensions, including domain, evidence type, and tool requirements, to support fine-grained failure analysis.

4.2 Baselines and Configurations

Fairness principle and execution contract. For Scenario 1, all architecture configurations share the same LLM (Gemini 2.5 Flash Lite, $T=0.0$), knowledge representations, retrieval

stack (BM25 + dense + RRF + bge-reranker-v2-m3, $k=7$), and deterministic calculator access. The intended contrast is the *orchestration architecture*. Scenario 2 intentionally changes one component at a time. To reduce retrieval variance in Scenario 1, all three architecture variants evaluate over the identical shared full-stack evidence retrieval output, yielding matched pre-generation Source Recall (78.6%) and Source AP (70.8%). The shared evidence context excludes outputs of tools whose invocation is evaluated; tool-derived results are available only after successful tool execution. Differences in final answers may be associated with the combined changes in tool privilege scoping, prompt specialization, and agent routing; Scenario 1 does not isolate their individual effects. Multi-agent variants use intent-routed scoped registries with identical execution bounds: maximum 3 tool calls, matching context limits, 2,048 output-token budget, and 30-second timeouts. Token accounting for CTU-CHAT aggregates all pipeline invocations (supervisor routing, contract repair, worker generation, and retries) for comparison with the single-agent baseline.

Scenario 1 — Architecture Comparison (RQ1): Three configurations are evaluated on the held-out set ($N=100$ queries, 300 runs under greedy decoding $T=0.0$): **S1-A:** Unified Single Agent with all 11 tools, no supervisor or routing; **S1-B:** Routed Generic Worker—supervisor routes to domain workers with scoped tools but a single generic prompt; **S1-C:** Full CTU-Chat with supervisor, contract repair, specialist prompts, scoped registries, and deterministic computation.

Scenario 2 — Ablation (RQ2): Five ablations each remove one component from Full CTU-Chat ($N=100$ queries per ablation, $T=0.0$): **S2-A1:** No Specialist Policy (generic worker prompt); **S2-A2:** No Tool Isolation (all 11 tools visible); **S2-A3:** Uniform Text-RAG Baseline (graph data serialized to text, traversal disabled); **S2-A4:** No Deterministic Calculation (LLM generates arithmetic directly); **S2-A5:** No Route Repair (raw supervisor output used directly).

Scenario 3 — Tool Ambiguity (RQ3): Fifty fixed cases are evaluated while 40 semantic-neighbor tools expand registries to 11, 31, and 51 tools. Four configurations are compared: Full-Registry Single Agent, Global Top-10 Single Agent, Routed Generic, and Routed Specialist. Each case uses 3 counterbalanced tool-order blocks, yielding $50 \times 4 \times 3 \times 3 = 1,800$ decisions across counterbalanced blocks.

4.3 Evaluation Metrics

Metrics span four groups. *Final-answer quality:* Required-Fact Coverage (continuous proportion of reference facts verified in generated response), Answer Correctness, Answer Relevancy, and Faithfulness (Ragas [4]). *Evidence quality:* Source Recall, Source AP, Context Recall, Context Precision. *Execution reliability:* Tool Invocations per Query, Tool Selection Accuracy, Argument EM/Contract Match, Result Accuracy, and route repair rate. *Operational cost:* input/output tokens, LLM call counts, and wall-clock latency. In comparative evaluation, E2E Success Rate serves as the primary endpoint. Key secondary endpoints are Required-Fact Coverage, tool invocations, token cost, and

latency. Source Recall and Source AP serve as controlled retrieval diagnostics, expected identical by design across Scenario 1 configurations.

E2E Success definition. For retrieval queries, E2E Success is evaluated under the criterion of required facts $\geq 50\%$, absence of unsupported claims, and retrieval of verified sources, complemented by continuous Fact Coverage. For tool queries: correct route, tool, arguments, result, and answer; for the Unified Single Agent baseline, routing correctness is evaluated as N/A by construction as it does not employ a router. For adversarial queries: absence of unnecessary tool calls and appropriate clarification or refusal. For the arithmetic subset ($N=9$), numeric exact match against official bursar schedules is independently audited.

4.4 Statistical Protocol

Evaluations use greedy decoding ($T=0.0$) to minimize sampling variance. Paired differences across configurations are computed at the query level ($N=100$); 95% CIs are estimated using 10,000 bootstrap resamples. The primary confirmatory hypothesis is CTU-Chat versus Unified Single Agent on E2E Success. Generic-worker comparisons, continuous Fact Coverage, token cost, and latency are evaluated as secondary outcomes with reported effect sizes and 95% bootstrap CIs. Scenario 3 degradation is computed as $\Delta = \text{score}_{51} - \text{score}_{11}$ across counterbalanced tool-order blocks.

4.5 Infrastructure

Gemini 2.5 Flash Lite via Vertex AI ($T=0.0$). Neural reranking on a GTX 1650 (evaluation hardware). Neo4j 5.20 stores relational curricula; PostgreSQL stores parent chunks (2,800-char, 100 overlap); Qdrant indexes child chunks (400-char, 50 overlap) with a 768-dimensional vietnamese-bi-encoder. System uses bge-reranker-v2-m3, LangGraph StateGraph, and Redis 7. Latency excludes the first warm-up run. Prompt templates, tool contracts, and graph schema are archived.

5 Results

5.1 Scenario 1 — Architecture-Level Comparison

Table 1 compares the three configurations on the 100-query held-out benchmark. Because pre-generation evidence retrieval is held fixed across configurations, Source Recall (78.6%) and Source AP (70.8%) are identical across all three architectures. The observed differences in final answer quality and operational cost may be associated with the combined orchestration changes described in Section 4.

Pairwise bootstrap analysis (10,000 resamples): (1) CTU-Chat vs. Single Agent: CTU-Chat achieves a 10.0 pp higher E2E point estimate (76.0% vs. 66.0%, paired 95% bootstrap CI [+2.0, +19.0] pp; exact McNemar $p=0.0414$), alongside higher Fact Coverage ($\Delta=+3.58$ pp, 95% CI [−1.2, +8.5] pp) and 41.4% fewer input tokens alongside 26.1% fewer total tokens ($\Delta= - 2,044$ total tokens, CI [−2,710, −1,378]; Single: 7,826 vs. CTU: 5,782), at the cost of +5.49 s mean end-to-end latency. (2) CTU-Chat vs.

Table 1. Scenario 1: Architecture-level comparison ($N = 100$). Tools/Q = mean tool invocations (↓). Bold = best reliability / lowest cost.

Arch.	E2E	Fact	Src.R	Src.AP	Tools/Q	In Tok	Out Tok	Lat. (ms)
Single Agent	66.0	57.5	78.6	70.8	0.41	6296	1530	5159
Routed Gen.	66.0	54.2	78.6	70.8	0.31	3473	1415	8921
CTU-Chat	76.0	61.0	78.6	70.8	0.28	3690	2092	10652

Routed Generic: specialist policies raise Fact Coverage by +6.85 pp (CI [+0.7, +13.0] pp, $p < 0.05$). (3) Routed Generic vs. Single Agent: comparable token savings ($\Delta = -2,824$ input tokens) but a -3.27 pp Fact Coverage point estimate (CI $[-9.0, +2.4]$ pp); the interval includes zero.

On the 20 cross-domain queries, CTU-Chat achieves an 80.0% E2E point estimate compared with 70.0% (Single) and 65.0% (Generic). A sensitivity check across fact-coverage thresholds shows that CTU-Chat has a higher point estimate at $\theta=0.50$ (76.0% vs. 66.0%; +10.0 pp) and $\theta=0.60$ (54.0% vs. 50.0%; +4.0 pp), while Single Agent has a slightly higher point estimate at $\theta=0.70$ (30.0% vs. 27.0%), $\theta=0.75$ (30.0% vs. 27.0%), and $\theta=1.00$ (25.0% vs. 23.0%).

Latency decomposition: The +5,493 ms mean latency difference between CTU-CHAT (10,652 ms) and Single Agent (5,159 ms) is attributable to two sequential processing stages. First, supervisor dispatch requires an extra round-trip (2.00 vs. 1.34 mean LLM calls per query), which accounts for +3,762 ms in the intermediate Routed Generic configuration (8,921 ms). Second, specialist contract compliance and explicit chain-of-evidence formatting generate +677 additional output tokens (2,092 vs. 1,415), contributing the remaining +1,731 ms. This generation-phase effect is corroborated by ablation S2-A4 (Section 5.2), where omitting structured calculations and strict contract schemas reduces latency by 4,540 ms.

5.2 Scenario 2 — Architectural Ablation

Table 2 removes one mechanism at a time from Full CTU-Chat ($N=100$). Deltas are relative to the full system.

The two largest observed degradations occur in *No Specialist Policy* ($\Delta E2E = -10.0$ pp, 95% CI $[-19.0, -1.0]$ pp; $\Delta \text{Fact} = -6.85$ pp, CI $[-13.0, -0.7]$ pp) and *Uniform Text-RAG Baseline* ($\Delta E2E = -9.0$ pp, CI $[-15.0, -4.0]$ pp; $\Delta \text{Src.R} = -12.3$ pp, CI $[-16.5, -8.3]$ pp). Removing tool isolation causes no change in point estimate ($\Delta E2E = 0.0$ pp, CI $[-6.0, +6.0]$ pp) while expanding input tokens by +1,270. Disabling route repair yields $\Delta E2E = -2.0$ pp (CI $[-7.0, +3.0]$ pp), disproportionately harming cross-domain queries (-15.0 pp, dropping from 80.0% to 65.0%). Disabling deterministic calculation produces $\Delta E2E = -4.0$ pp (CI $[-9.0, 0.0]$ pp) across the benchmark; on the arithmetic tuition subset ($N=9$), exact-match answers drop from 8/9 to 3/9 (-55.6 pp exact match pass) due to unconstrained arithmetic hallucination. Collectively, the ablations distinguish two functional roles: specialist policies, heterogeneous evidence, and deterministic calculation drive accuracy at the current 11-tool scale, while tool isolation (S2-A2) functions as an efficiency mechanism (+1,270 input tokens, $\Delta E2E = 0.0$ pp) whose

Table 2. Scenario 2: Architectural ablation results ($N = 100$). Panel A reports overall benchmark differences (Δ relative to Full CTU-CHAT). Panel B reports E2E Success (%) across query families.

Panel A: Overall Benchmark Ablations					
Ablated Component	ΔE2E	ΔFact	ΔSrc.R	ΔTok	ΔLat
No Spec. Policy	−10.0	−6.85	0.0	−217	−1731
Full Tool Vis.	0.0	+0.93	0.0	+1270	+296
Uniform Text-RAG	−9.0	−2.18	−12.3	−300	−1910
No Route Rep.	−2.0	−1.27	0.0	−219	−1610
No Determ. Calc. [†]	−4.0	−0.93	0.0	−782	−4540

Panel B: E2E Success (%) by Query Family				
Configuration	Graph ($n=19$)	Struct-fin ($n=28$)	Narr-reg ($n=33$)	Cross ($n=20$)
Full CTU-Chat	63.2	75.0	81.8	80.0
No Spec. Policy	52.6	67.9	72.7	65.0
Full Tool Vis.	68.4	67.9	84.8	80.0
Uniform Text-RAG	63.2	60.7	78.8	60.0
No Route Rep.	68.4	78.6	78.8	65.0

[†]Full benchmark with calculators disabled. Δ Tok/Lat in tokens/ms per query.

accuracy contribution is examined under larger registry scales in Scenario 3 (Section 5.3). Query family breakdown in Panel B confirms that heterogeneous evidence and specialist routing particularly bolster structured financial and cross-domain questions.

5.3 Scenario 3 — Tool Ambiguity Stress Test

Table 3 reports the semantic-neighbor experiment across 11–51 tools ($N=50$, 3 counter-balanced tool-order blocks, 1,800 primary decisions).

As the registry scales from 11 to 51 tools, the Full-Registry Single Agent changes from 86.0% to 83.3% E2E while input tokens increase by 124%. The Global Top-10 changes from 83.3% to 78.7% E2E, alongside a Gold Recall change from 100.0% to 96.0%. The Routed Specialist records 88.0% E2E at 51 tools with 9.8 visible tools. At the 51-tool endpoint, Routed Specialist has a +9.3 pp E2E difference over Top-10 Single Agent (95% CI [+2.0, +17.3] pp) and a +4.7 pp point-estimate difference over Full-Registry Single Agent (95% CI [−1.3, +11.3] pp) with 52.4% fewer input tokens (1,770 vs. 3,722). The degradation interaction (Specialist vs. Top-10 high-minus-low difference) is +7.3 pp (95% CI [0.0, +15.3] pp).

Analysis of stage-level timing across the 1,960 decisions reveals that Routed Specialist maintains a consistent latency profile across registry expansions (2,091 ms at 11 tools vs. 2,116 ms at 51 tools). Here, supervisor routing introduces a fixed overhead of 946.5–961.8 ms (~45% of total decision time), tool gating requires 1,051.6–1,079.3 ms, and tool execution takes 71.0–92.8 ms. In contrast, while the Full-Registry Single Agent avoids supervisor dispatch (0 ms routing overhead), its gating latency increases from 1,098.1 ms to 1,272.2 ms as prompt schema tokens scale from 1,663 to 3,722, demonstrating that the fixed routing overhead of the multi-agent architecture buys sub-linear input token scaling and resistance to semantic interference.

Table 3. Scenario 3: Semantic-neighbor tool ambiguity ($N = 50$, 3 counterbalanced tool-order blocks).

Reg.	Config.	Vis.	Gold-R	Sel.	E2E	Tok	Lat
11	Full-reg single	11	–	94.0	86.0	1663	1098
	Top-10 single	10	100.0	93.3	83.3	1541	1148
	Routed gen.	5.1	98.0	90.7	80.0	1204	1889
	Routed spec.	5.1	98.0	93.3	85.3	1441	1983
31	Full-reg single	31	–	91.3	84.7	2706	1157
	Top-10 single	10	98.0	90.0	81.3	1149	1033
	Routed gen.	9.8	98.0	88.0	79.3	1396	1958
	Routed spec.	9.8	98.0	93.3	88.7	1830	2015
51	Full-reg single	51	–	92.7	83.3	3722	1289
	Top-10 single	10	96.0	86.7	78.7	1036	997
	Routed gen.	9.8	98.0	91.3	82.7	1340	1907
	Routed spec.	9.8	98.0	92.7	88.0	1770	2021

5.4 Supporting Retrieval Validation

Table 4 validates the retrieval subsystem independently.

Table 4. Supporting evaluation: Progressive retrieval component stacking ($N = 100$).

Config.	Hit@1	Hit@3	P@5	R@5	MRR@10	Lat. (ms)
BM25 Lexical	0.530	0.770	0.206	0.617	0.669	7
Dense Semantic	0.460	0.650	0.176	0.524	0.560	34
Hybrid RRF	0.520	0.790	0.224	0.662	0.673	38
Hybrid + Reranker	0.700	0.910	0.254	0.751	0.805	15,366
Full Retrieval	0.740	0.920	0.302	0.861	0.833	21,858

BM25 records 0.530 Hit@1 and 0.669 MRR@10, compared with 0.460 Hit@1 and 0.560 MRR@10 for dense semantic search. Hybrid RRF achieves 0.790 Hit@3 and 0.520 Hit@1, balancing lexical precision and semantic recall. Adding the cross-encoder reranker (bge-reranker-v2-m3) yields the most pronounced individual retrieval gain, raising Hit@1 from 0.520 to 0.700 (+18.0 pp) and MRR@10 to 0.805 at the expense of reranker inference latency. The full retrieval configuration reaches 0.740 Hit@1, 0.861 Recall@5, and 0.833 MRR@10. Because this final configuration incorporates domain-specific evidence components, the difference cannot be attributed to one component alone.

5.5 Failure Analysis

Table 5 categorizes error modes across the 46 failure instances ($< 50\%$ Fact Coverage) in Scenario 1 (27 for Single Agent, 19 for CTU-Chat).

Table 5. Failure taxonomy across architecture configurations ($N = 100$). Categories are non-mutually exclusive; a single query may exhibit multiple error modes.

Category	Failure Type	Single CTU	
Routing	Misroute corrected by repair	N/A	4
	Unrecoverable misroute	N/A	2
Retrieval	Gold source absent	2	2
	Cross-domain partial omission	5	3
	Reranker demoted gold below $k=5$	4	4
Tool	Invalid/default arguments	4	1
	Wrong-family hallucination	3	1
	Unnecessary invocation	5	2
Generation	Tool result ignored in text	2	1
	Partial fact omission	11	9
Total	Fact Coverage < 50%	27	19

CTU-Chat has 2 recorded tool errors versus 7 for Single Agent and 2 unnecessary calls versus 5. Gold-source absence and reranker-demotion counts are identical (2 and 4 respectively), while cross-domain partial omissions number 3 versus 5.

6 Discussion

6.1 Architectural Findings and Operational Trade-offs

RQ1 (Architecture-Level Effectiveness): The architecture-level comparison indicates that supervisor-routed multi-agent decomposition achieves a 10.0 pp higher E2E point estimate than the Unified Single Agent (76.0% vs. 66.0%; paired bootstrap CI [+2.0, +19.0] pp; exact McNemar $p=0.0414$), alongside a +3.58 pp Fact Coverage advantage (95% CI [-1.2, +8.5] pp) and 41.4% fewer input tokens (3,690 vs. 6,296), at the expense of higher wall-clock latency (10,652 vs. 5,159 ms). Routed Generic records lower Fact Coverage than CTU-CHAT (54.2% vs. 61.0%; $\Delta = -6.85$ pp, $p < 0.05$), indicating that specialist policies contribute substantially to answer fidelity under shared routing.

RQ2 (Mechanisms and Tool Isolation Reframe): Component ablations reveal two distinct functional roles. *Accuracy drivers:* Heterogeneous evidence access (S2-A3) yields the largest Source Recall drop (-12.3 pp, including -17.8 pp on structured finance); specialist prompt policies (S2-A1) drive Fact Coverage (-6.85 pp); route repair (S2-A5) provides cross-domain resilience (-15.0 pp); and deterministic calculators (S2-A4) prevent arithmetic hallucination (-55.6 pp exact-match on tuition). Conversely, *efficiency and scalability mechanisms:* Tool isolation (S2-A2) does not alter the E2E point estimate at the baseline 11-tool scale ($\Delta E2E = 0.0$ pp, CI [-6.0, +6.0] pp) while expanding input tokens by +1,270 (+34.4%). This confirms that at small scales, tool isolation functions primarily as a token-efficiency and cost-containment safeguard rather than an accuracy driver, as the base LLM comfortably handles 11 schemas without severe attention distraction. The essential value of tool isolation emerges under registry expansion. In

Scenario 3, when scaling to 51 tools with semantic distractors, the scoped specialist maintains 88.0% E2E with 9.8 visible tools, compared with 83.3% for the Full-Registry Single Agent (+4.7 pp, CI $[-1.3, +11.3]$ pp) and 78.7% for Global Top-10 (+9.3 pp, CI $[+2.0, +17.3]$ pp), while saving 52.4% input tokens (1,770 vs. 3,722). Tool isolation thus functions as a preventive architectural boundary that limits context expansion and insulates agents against semantic interference as services grow.

RQ3 (Tool Ambiguity and Stress Testing): Under controlled semantic-neighbor injection across 11–51 tools, the Routed Specialist sustains 88.0% E2E at 51 tools, whereas Global Top-10 degrades to 78.7% due to candidate shortlist truncation (Gold Recall dropping to 96.0%). The Full-Registry Single Agent incurs a 124% input token penalty to achieve 83.3% E2E. Specialization effectively decouples per-agent context load from aggregate registry size.

Latency Decomposition and Trade-offs: The empirical data resolve where latency is expended across architectures. (1) *Sequential dispatch overhead:* Supervisor routing adds a sequential LLM hop (~ 3.8 s in Scenario 1; ~ 0.95 s in Scenario 3) absent in monolithic systems, making single agents faster when tool pools are compact. (2) *Generation versus dispatch latency:* Latency decomposition demonstrates that the majority of CTU-CHAT’s duration is spent in response generation (~ 6.8 s), where specialist contracts enforce structured evidence citation and reasoning, generating 48% more output tokens (2,092 vs. 1,415). Ablation S2-A4 confirms that removing structured calculation and contract formatting reduces latency by 4,540 ms, though compromising calculation exactness. (3) *Operational guidance:* Monolithic single-agent designs remain preferable for low-latency, single-domain workflows with small tool pools (≤ 10); supervisor-routed multi-agent designs are justified when institutional counseling demands heterogeneous evidence, calculation fidelity, and scalable tool registries.

6.2 Relation to Prior Work

While ReAct [20] and Toolformer [16] establish tool use within single-agent paradigms, our results indicate that full-registry single agents suffer substantial prompt expansion as tools accumulate. Unlike collaborative multi-agent frameworks (e.g., AutoGen [18]) where unstructured chatter compounds token consumption and latency, CTU-CHAT enforces a bounded, single-owner dispatch model with deterministic route repair. The S2-A3 ablation highlights the fragility of uniform text retrieval on structured relational queries, corroborating the rationale for heterogeneous evidence representations in GraphRAG [3]. Furthermore, whereas Adaptive-RAG [6] dynamically branches based on linguistic complexity, CTU-CHAT routes queries according to institutional data modalities.

6.3 Threats to Validity

Internal: All primary evaluations utilize Gemini 2.5 Flash Lite with greedy decoding ($\tau=0$) to reduce sampling variance; API serving latency and backend scheduling variations remain potential operational confounds. **External:** The benchmark is grounded in statutes and curricula from Can Tho University; validation across multi-campus institutions with bilingual or trimester policies remains future work. Semantic-neighbor tools in Scenario 3 are controlled synthetic constructs modeling intra-domain ambiguity. **Construct:** Metric

scoring incorporates automated evaluation via Ragas alongside deterministic exact-match checks; human verification was conducted on a subset to validate metric concordance. **Statistical:** Evaluations are based on 100 held-out cases with paired bootstrap resampling (10,000 iterations); the arithmetic sub-cohort ($N=9$) has a small sample size, though the difference in exact match (8/9 vs. 3/9) is substantial. **Operational:** Latency measurements reflect cloud-hosted API execution and public network connectivity.

7 Conclusion and Future Work

This study evaluated CTU-CHAT, a supervisor-routed multi-agent RAG framework for university counseling, across architecture comparison, ablation, and tool ambiguity stress testing. Three principal findings address the research questions: (1) CTU-Chat achieves a 10.0 pp higher End-to-End Success point estimate than a matched single-agent baseline (76.0% vs. 66.0%; paired CI [+2.0, +19.0] pp; exact McNemar $p=0.0414$) with 41.4% fewer input tokens; (2) removing specialist policies corresponds to a −10.0 pp E2E drop, while replacing heterogeneous evidence allocation reduces Source Recall by −12.3 pp; and (3) the Routed Specialist sustains 88.0% E2E at 51 tools, compared with 83.3% for the Full-Registry Single Agent and 78.7% for Global Top-10. Supervisor-routed specialization effectively supports heterogeneous institutional counseling: domain policies, representation-specific evidence paths, and deterministic calculations drive reliability, while scoped tool access provides a crucial scalability safeguard under registry expansion, trading off against sequential dispatch latency. Future work includes reducing dispatch latency via speculative routing, evaluating multi-institutional transfer across bilingual curricula, and conducting live longitudinal counseling trials.

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References

1. Can Tho University: Can tho university inaugurates the 2025–2026 academic year (2025), <https://www.ctu.edu.vn>, accessed 13 September 2026
2. Cormack, G.V., Clarke, C.L.A., Buettcher, S.: Reciprocal rank fusion outperforms condorcet and individual rank learning methods. In: Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval. pp. 758–759 (2009). <https://doi.org/10.1145/1571941.1572114>
3. Edge, D., Trinh, H., Cheng, N., Bradley, J., Chao, A., Mody, A., Truitt, S., Larson, J.: From local to global: A graph RAG approach to query-focused summarization. arXiv preprint arXiv:2404.16130 (2024)
4. Es, S., James, J., Espinosa-Anke, L., Schockaert, S.: RAGAS: Automated evaluation of retrieval augmented generation. In: Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations. pp. 150–158 (2024). <https://doi.org/10.18653/v1/2024.eacl-demo.16>
5. Gao, Y., Xiong, Y., Gao, X., Jia, K., Pan, J., Bi, Y., Dai, Y., Sun, J., Wang, Q., Wang, H.: Retrieval-augmented generation for large language models: A survey. arXiv preprint arXiv:2312.10997 (2023)

6. Jeong, S., Baek, J., Cho, S., Hwang, S.J., Park, J.C.: Adaptive-RAG: Learning to adapt retrieval-augmented large language models through question complexity. In: Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies. pp. 7036–7050 (2024). <https://doi.org/10.18653/v1/2024.naacl-long.389>
7. Karpukhin, V., Oğuz, B., Min, S., Lewis, P., Wu, L., Edunov, S., Chen, D., Yih, W.t.: Dense passage retrieval for open-domain question answering. In: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP). pp. 6769–6781 (2020). <https://doi.org/10.18653/v1/2020.emnlp-main.550>
8. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Kuttler, H., Lewis, M., Yih, W.t., Rocktäschel, T., Riedel, S., Kiela, D.: Retrieval-augmented generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems* **33**, 9459–9474 (2020)
9. Ma, T., La, T.T., Le Huu, L.T., Nguyen, M.N., Pham Luu, K.V.: REBot: From RAG to CatRAG with semantic enrichment and graph routing. *arXiv preprint arXiv:2510.01800* (2025)
10. Ma, T.T., Chau, T.K., Nguyen, T.P., Huynh, G.K., Nguyen, T.H., Tran, V.C., Do, T.N.: CAAS: Ho tro tu van tuyen sinh voi Chat-Voice su dung ChatGPT va may hoc. In: Proceedings of the 16th National Conference on Fundamental and Applied Information Technology Research (FAIR). pp. 526–533 (2023). <https://doi.org/10.15625/vap.2023.0079>
11. Nguyen, D.Q., Nguyen, A.T.: PhoBERT: Pre-trained language models for Vietnamese. In: Findings of the Association for Computational Linguistics: EMNLP 2020. pp. 1037–1042 (2020). <https://doi.org/10.18653/v1/2020.findings-emnlp.92>
12. Nogueira, R., Cho, K.: Passage re-ranking with BERT. *arXiv preprint arXiv:1901.04085* (2019)
13. Pan, S., Luo, L., Wang, Y., Chen, C., Wang, J., Wu, X.: Unifying large language models and knowledge graphs: A roadmap. *IEEE Transactions on Knowledge and Data Engineering* **36**(7), 3580–3599 (2024). <https://doi.org/10.1109/TKDE.2024.3352100>
14. Qian, J., Wang, H., Li, Z., Li, S., Yan, X.: Limitations of language models in arithmetic and symbolic induction. In: Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics. pp. 9285–9298 (2023). <https://doi.org/10.18653/v1/2023.acl-long.516>
15. Robertson, S., Zaragoza, H.: The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval* **3**(4), 333–389 (2009). <https://doi.org/10.1561/15000000019>
16. Schick, T., Dwivedi-Yu, J., Dessì, R., Raileanu, R., Lomeli, M., Hambro, E., Zettlemoyer, L., Cancedda, N., Scialom, T.: Toolformer: Language models can teach themselves to use tools. In: *Advances in Neural Information Processing Systems*. vol. 36 (2023). <https://doi.org/10.52202/075280-2997>
17. Wollny, S., Schneider, J., Di Mitri, D., Weidlich, J., Rittberger, M., Drachsler, H.: Are we there yet? a systematic literature review on chatbots in education. *Frontiers in Artificial Intelligence* **4**, 654924 (2021). <https://doi.org/10.3389/frai.2021.654924>
18. Wu, Q., Bansal, G., Zhang, J., Wu, Y., Li, B., Zhu, E., Jiang, L., Zhang, X., Zhang, C., Liu, J., et al.: AutoGen: Enabling next-gen LLM applications via multi-agent conversation. *arXiv preprint arXiv:2308.08155* (2023)
19. Xiao, S., Liu, Z., Zhang, P., Muennighoff, N.: C-Pack: Packaged resources to advance general chinese and cross-lingual embedding. *arXiv preprint arXiv:2309.07597* (2023)
20. Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., Cao, Y.: ReAct: Synergizing reasoning and acting in language models. In: *International Conference on Learning Representations (ICLR)* (2023)